

Conjecture on k-Almost-Prime Hierarchy in Consecutive Square Intervals

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Conjecture Statement

Conjecture 3.1: Basic Diversity Conjecture

Informal Statement:

For all sufficiently large x , the interval $[x^2, (x+1)^2]$ contains at least one prime, one semiprime (2-almost-prime), and one 3-almost-prime.

Formal Statement:

$$\exists x_0 \in \mathbb{N} : \forall x \geq x_0,$$

$$\exists p, s, t \in [x^2, (x+1)^2] \cap \mathbb{N} \text{ such that:}$$

- p is prime (1-almost-prime)
 - $s = p_1 \cdot p_2$ where p_1, p_2 are prime (2-almost-prime)
 - $t = q_1 \cdot q_2 \cdot q_3$ where q_1, q_2, q_3 are prime (3-almost-prime)
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Conjecture 3.2: Generic Hierarchy Conjecture

Informal Statement:

For all sufficiently large x , unless the interval contains a large prime power, it contains the complete hierarchy of k -almost-primes for small k .

Formal Statement:

$$\forall K \in \mathbb{N}, \exists x_0(K) \in \mathbb{N} : \forall x \geq x_0(K),$$

If the interval $[x^2, (x+1)^2]$ contains no prime power p^m with $m \geq 4$, then:

$$\forall k \in \{1, 2, 3, \dots, K\}, \exists n \in [x^2, (x+1)^2] \text{ such that } \Omega(n) = k$$

where $\Omega(n)$ denotes the number of prime factors of n counted with multiplicity.

Conjecture 3.3: Density of Complete Hierarchies

Informal Statement:

As x grows, the proportion of intervals containing the complete k -almost-prime hierarchy approaches 1.

Formal Statement:

$$\forall K \in \mathbb{N}, \lim_{X \rightarrow \infty} \frac{1}{X} |\{x \leq X : [x^2, (x+1)^2] \text{ contains all } k\text{-almost-primes for } k \leq K\}| = 1$$

Context and Related Work

Definitions

A **k -almost-prime** (or k -semiprime) is a natural number with exactly k prime factors counted with multiplicity. Formally:

$$n \text{ is } k\text{-almost-prime} \iff \Omega(n) = k$$

Examples:

- 1-almost-primes: 2, 3, 5, 7, 11, 13, ... (the primes)
- 2-almost-primes: 4, 6, 9, 10, 14, 15, 21, 22, ... (semiprimes)
- 3-almost-primes: 8, 12, 18, 20, 27, 28, 30, ...

Known Results

1. **Legendre's Conjecture (1808):** For every $n \geq 1$, there exists at least one prime in the interval $[n^2, (n+1)^2]$.
 - Status: OPEN (one of Landau's four problems, 1912)
 - Even assuming RH, this remains unproven
 2. **Ingham (1937):** Proved there is always a prime in $[n^3, (n+1)^3]$ for sufficiently large n .
 3. **Baker-Harman-Pintz (2001):** There is always a prime in $[x - x^{0.525}, x]$ for sufficiently large x .
 4. **Distribution of k -almost-primes:** The number of k -almost-primes up to x is asymptotically: $\pi_k(x) \sim \frac{x (\log \log x)^{k-1}}{(k-1)! \log x}$
 5. **Chen's Theorem (1973):** There are infinitely many primes p such that $p+2$ is either prime or semiprime.
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Distinction from Related Problems

Classical Approach: Existence of Primes

- Legendre, Brocard, Oppermann: Focus on existence of primes in intervals
- Ask: "How many primes are in this interval?"

Proposed Approach: Diversity of Factorization Types

- Focus on **variety** of factorization structures
- Ask: "Does the interval contain representatives from different k-almost-prime classes?"

Key Differences:

1. **Diversity vs. Quantity:** We don't ask for many primes, but for variety in factorization types.
 2. **Structural Question:** Tests whether short intervals are "rich enough" to contain different arithmetic structures.
 3. **Obstruction Identification:** Prime powers create predictable gaps in the hierarchy.
 4. **Density Perspective:** Asks about typical behavior vs. exceptional cases.
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Significance

If True:

These conjectures would establish that:

1. **Factorization Diversity:** Short intervals between consecutive squares contain diverse factorization types, not just concentrated in one class.
2. **Generic Completeness:** Except for rare, structured exceptions (prime powers), intervals contain the full k-almost-prime hierarchy.
3. **Predictable Obstructions:** Prime powers are the primary (only?) cause of missing levels in the hierarchy.
4. **Density Result:** The exceptional cases (missing hierarchy levels) have density approaching 0.

If False:

A counterexample would reveal:

1. **Hidden Structure:** There exist obstructions beyond prime powers preventing complete hierarchies.
 2. **Gap Phenomenon:** Certain k-levels are systematically absent even without prime power obstructions.
 3. **Density Limitations:** The proportion of intervals with complete hierarchies may be bounded away from 1.
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Relationship to Classical Conjectures

Connection to Legendre's Conjecture

Conjecture 3.1 implies Legendre's Conjecture (for large x):

- If every interval $[x^2, (x + 1)^2]$ contains a prime (among other types), then Legendre holds.
- Therefore, Conjecture 3.1 is at least as hard as one of Landau's problems.

However:

- The semiprime and 3-almost-prime requirements are likely easier (these are more abundant).
- The prime requirement is the bottleneck.

Comparison to Brocard's Conjecture

Brocard: There are at least 4 primes between n^2 and $(n + 1)^2$.

Our approach: Instead of multiple primes, we ask for diversity:

- Orthogonal questions (quantity vs. variety)
 - Both could be true independently
 - Brocard is likely harder (requires 4 primes, not just 1)
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Computational Observations

Empirical Findings

Through computational investigation for $x \leq 10^6$, the following patterns emerge:

1. **Generic Behavior:** Most intervals contain the complete hierarchy for $k \leq 6$.
2. **Prime Power Obstructions:** When levels are missing, the interval typically contains a prime power p^m with $m \geq 4$.
 - Example: If $2^{10} = 1024 \in [x^2, (x + 1)^2]$, then intermediate levels (e.g., $k = 4, 5, 6, \dots$) may be absent.
3. **Rare Gaps:** Intervals missing hierarchy levels become increasingly rare as x grows.
4. **Typical Missing Levels:** When gaps occur, they typically affect mid-range k (around $k = 4$ to $k = 8$), not the small values $k \in \{1, 2, 3\}$.

Examples of Exceptional Intervals

Intervals containing large prime powers:

- If $[x^2, (x + 1)^2]$ contains 2^{15} , 3^{10} , or 5^8 , the hierarchy often has gaps.

- The presence of p^m with large m "fills" the k -almost-prime role for $k = m$, but leaves gaps at smaller k values.

Example:

- Suppose $2^8 = 256 \in [x^2, (x+1)^2]$.
 - This is an 8-almost-prime.
 - The interval might lack 4-almost-primes, 5-almost-primes, etc., but contain this single 8-almost-prime.
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Open Questions

1. **Explicit Bound for x_0 :** Can we determine the smallest x_0 such that Conjecture 3.1 holds for all $x \geq x_0$?
 2. **Prime Power Characterization:** Is every missing level in the hierarchy explained by the presence of a large prime power, or are there other obstructions?
 3. **Asymptotic Density:** Can we prove that the density of intervals with complete k -hierarchies approaches 1, or find an explicit rate of convergence?
 4. **Maximum Hierarchy Depth:** For a given x , what is the largest $K(x)$ such that the interval contains all k -almost-primes for $k \leq K(x)$?
 5. **Conditional Results:** Can we prove Conjectures 3.2 or 3.3 assuming Legendre's Conjecture or the Riemann Hypothesis?
 6. **Lower Bounds on Abundance:** Can we prove that the interval contains at least $c_k(x)$ distinct k -almost-primes for each k , where $c_k(x) \rightarrow \infty$?
 7. **Generalization to Other Intervals:** Do similar hierarchy results hold for intervals like $[x^a, (x+1)^a]$ for $a \geq 3$, or $[x, x+x^\theta]$ for fixed $\theta \in (0, 1)$?
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Methodology and Approach

Potential approaches to these conjectures:

1. Heuristic Density Arguments

The interval $[x^2, (x+1)^2]$ has length approximately $2x$.

Expected counts:

- Primes: $\sim \frac{2x}{\log(x^2)} = \frac{x}{\log x}$ (many for large x)
- 2-almost-primes: $\sim \frac{2x \log \log x}{\log x}$ (even more abundant)
- k -almost-primes: $\sim \frac{2x (\log \log x)^{k-1}}{(k-1)! \log x}$ (very abundant)

Heuristically, there should be many representatives of each type.

2. Sieve Methods

- Use combinatorial sieve techniques to control the distribution of k-almost-primes.
- Adaptation of Chen's sieve or GPY (Goldston-Pintz-Yıldırım) methods might be relevant.

3. Conditional Proofs

- **Assuming Legendre:** Prove the semiprime and 3-almost-prime parts.
- **Assuming RH:** Strengthen density estimates for k-almost-primes in short intervals.

4. Computational Verification

- Verify conjectures for $x \leq 10^9$ or beyond.
- Classify exceptional cases (intervals with missing levels).
- Identify patterns in which levels are missing and when.

5. Prime Power Analysis

- Prove that large prime powers are the primary obstruction.
- Show that if no p^m with $m \geq 4$ exists in the interval, then the hierarchy is complete.

Status

OPEN - As of January 2026, these conjectures appear to be original.

While the distribution of k-almost-primes is well-studied, the specific question of **hierarchy completeness** in consecutive square intervals has not been addressed in this form.

The approach of studying **diversity of factorization types** rather than abundance of a single type represents a novel perspective on short interval arithmetic.

Examples and Special Cases

Small Values of x

$x = 10$: Interval [100, 121]

- Primes: 101, 103, 107, 109, 113 ✓
- 2-almost-primes: $106 = 2 \cdot 53$, $111 = 3 \cdot 37$, $115 = 5 \cdot 23$, $118 = 2 \cdot 59$, $119 = 7 \cdot 17$, $121 = 11 \cdot 11$ ✓
- 3-almost-primes: $102 = 2 \cdot 3 \cdot 17$, $105 = 3 \cdot 5 \cdot 7$, $110 = 2 \cdot 5 \cdot 11$, $114 = 2 \cdot 3 \cdot 19$ ✓

Hierarchy levels 1, 2, 3 all present ✓

$x = 31$: Interval $[961, 1024]$

- Contains $2^{10} = 1024$ (10-almost-prime)
- Primes: 967, 971, 977, 983, 991, 997, 1009, 1013, 1019, 1021 ✓
- 2-almost-primes: $962 = 2 \cdot 481$, $965 = 5 \cdot 193$, ... ✓
- 3-almost-primes: $966 = 2 \cdot 3 \cdot 161$, $969 = 3 \cdot 17 \cdot 19$, ... ✓
- 4-almost-primes: ?
- 5-almost-primes: ?

Some mid-levels may be missing due to the presence of 2^{10} .

Acknowledgments

This conjecture emerged from computational exploration of prime and composite distribution in short intervals, motivated by classical questions about primes between consecutive squares. The observation of prime power obstructions arose from empirical investigation of exceptional cases where the k -almost-prime hierarchy appeared incomplete.

References

1. Legendre, A.-M. (1808). *Essai sur la théorie des nombres* (2nd ed.).
2. Landau, E. (1912). *Handbuch der Lehre von der Verteilung der Primzahlen*.
3. Ingham, A.E. (1937). "On the difference between consecutive primes." *Quarterly Journal of Mathematics*.
4. Chen, J. R. (1973). "On the representation of a large even integer as the sum of a prime and the product of at most two primes." *Sci. Sinica* 16.
5. Baker, R. C., Harman, G., & Pintz, J. (2001). "The difference between consecutive primes, II." *Proceedings of the London Mathematical Society*.
6. Landau, E. (1900). "Über die zahlentheoretische Funktion $\varphi(n)$." *Sitzungsber. Kaiserl. Akad. Wiss. Wien*.